

Supporting Information for:

Title: Model predicts coastal mussel toxicity from *in situ* phytoplankton concentrations

Authors: Sylvain Gaillard^{1*}, David K. Ralston¹, V. Monica Bricelj², Roman Marin III³, Bruce Keafer¹, Dennis J. McGillicuddy Jr.¹, Donald M. Anderson¹.

* Corresponding author sylvain.gllrd@gmail.com

¹ Woods Hole Oceanographic Institution, Woods Hole MA 02543 USA

² TechMar Research Inc., Brooklyn, NY, USA

³ Monterey Bay Aquarium Research Institute, Moss Landing, CA, USA

- 1 **Table S1.** Field or laboratory studies modeling paralytic shellfish toxin dynamics in commercially important mussel species (*Mytilus* spp.). SD:
- 2 standard deviation; n.d.: not determined; n.a.: not available. Summary values of uptake and depuration parameters are expressed as the averaged
- 3 mean variable values for each study (when data were available).

Bivalve mollusk species	Dinoflagellate species, exposure source and concentrations (cells L ⁻¹)	Type of model	Uptake parameter	Depuration parameter	Comments	Reference
<i>Mytilus edulis</i> (blue mussel)	<i>Alexandrium catenella</i> , bloom, 10 ⁵ to 10 ⁶	One-compartment or two-compartment	$1.1 \times 10^{-2} \pm 9.5 \times 10^{-3}$ (average $\pm$ SD) 9.0×10^{-4} (min) to 2.6×10^{-2} (max) $\mu\text{g STX eq. } 100 \text{ g}^{-1} \text{ tissue L cell}^{-1} \text{ d}^{-1}$ ^a	Constant: 0.24 and 0.40 d ⁻¹	One-compartment model provided a "useful representation of toxin dynamics", but the fit was "better" using a two-compartment model. The modeled accumulation was not "very sensitive" to modification of the depuration parameter.	Silvert and Cembella 1995

<i>M. edulis</i>	<i>A. catenella</i> , bloom, 1.0×10^2 (average) and 6.8×10^3 (max).	One-compartment	Variable: $3.1 \times 10^{-2} \pm 1.7 \times 10^{-2}$ (average $\pm$ SD) 9.0×10^{-3} (min) to 7.2×10^{-2} $\mu\text{g STX eq. } 100 \text{ g}^{-1}$ tissue L cell $^{-1}$ d $^{-1}$	Constant: 0.15 or variable: $0.16 \pm 9.3 \times 10^{-2}$ (average $\pm$ SD) 3.0×10^{-2} (min) to 0.35 (max)	Moderate to high R^2 depending on parameter values. Fit was sensitive to the uptake but not the depuration parameters and was higher when using locally fitted parameters.	This study
<i>M. galloprovincialis</i> (Mediterranean mussel)	<i>Gymnodinium catenatum</i> , bloom, n.d.	One or two-compartment	n.d.	Constant: 0.15 or 0.21 d $^{-1}$	Higher R^2 using two-compartment model (0.89) than one-compartment model (0.45). R^2 was sensitive to incorporation of cellular toxin concentration parameter and to low <i>A. catenella</i> concentrations.	Moroño and Blanco 1997
<i>M. galloprovincialis</i>	<i>G. catenatum</i> , bloom, n.d.	Two-compartment - depuration model	n.a.	Constant or variable: -8.0×10^{-2} (min) to 0.27 (max). Best fit: 0.15 d $^{-1}$	Two-compartment model provided a “better” fit than one-compartment model. “Better” fit when depuration was adjusted with environmental	Blanco et al. 1997

					(biotic and abiotic) variables. High R^2 of 0.92	
<i>M. galloprovincialis</i>	<i>A. minutum</i> , culture, 1×10^6	One or two-compartment	Constant: 4.6 L.h^{-1}	Constant: 0.15 or variable: 0 (min) to $0.28 \text{ (max) d}^{-1}$	One-compartment with constant uptake and depuration, and transformation of toxins, provided the “best” fit. Variation in depuration rate did not significantly increase the fit.	Blanco et al. 2003
<i>M. galloprovincialis</i>	<i>G. catenatum</i> , bloom and culture, $>10^3$	One-compartment - depuration model	n.a.	Fitted according to the toxin: 4.8×10^{-2} (min) to 8.0×10^{-2} (max) (<i>in vitro</i> exp) and 7.1×10^{-2} (min) to $0.15 \text{ (max) d}^{-1}$ (<i>in vivo</i> exp). Average: $8.0 \times 10^{-2} \text{ d}^{-1}$	Low R^2 . May be due to the “intraspecific variation” of toxin accumulation.	Botelho et al. 2010

<i>M. galloprovincialis</i>	<i>G. catenatum</i> , culture, 9.1×10^4 cells ⁻¹ d ⁻¹ individual ⁻¹	One-compartment	n.d.	Fitted according to the toxin, temperature, and pH: 6.0×10^{-2} to 0.90 d ⁻¹ . Average: 0.31 d ⁻¹	High R^2 averaging 0.84 . Lowest depuration rate in warm-acclimated (24 °C) animals vs. control (19 °C) (0.14 ± 0.11 vs. 0.26 ± 0.20 , average $\pm$ SD).	Braga et al. 2018
<i>M. coruscus</i> (Korean mussel)	<i>A. catenella</i> , culture, 5.6×10^3 to 1.0×10^7	One-compartment	$1.4 \times 10^{-5} \pm 1.0 \times 10^{-5}$ (average $\pm$ SD) 4.0×10^{-6} (min) to 3.6×10^{-5} (max) μ g STX eq. 100 g ⁻¹ tissue L cell ⁻¹ d ⁻¹ ^a	0.25 ± 0.15 (average $\pm$ SD) 6.0×10^{-2} (min) to 0.53 (max)	High R^2 averaging 0.89 . Depuration rates varied with temperature (0.18 ± 0.058 at control (24 °C) vs. 0.32 ± 0.18 at high temperature (30 °C)) and phytoplankton concentration (0.28 ± 0.14 at low vs. 0.22 ± 0.17 at high concentration).	Tang et al. 2021
Summary (average $\pm$ SD)			0.014 ± 0.016 μ g STX eq. 100 g ⁻¹ tissue L cell ⁻¹ d ⁻¹	0.16 ± 0.10 d ⁻¹		

4 ^a calculated from the authors data

Table S2. Environmental Sample Processor (ESP) deployment locations

Year	ESP	Latitude (dms)	Longitude (dms)
2014	ESP-A	43°10'36.486"N	70°26'35.610"W
	ESP-B	43°41'13.836"N	69°58'36.492"W
	ESP-C	43°48'10.536"N	69°27'26.946"W
2017	ESP-D	44°34'4.860"N	67°17'40.560"W
2019	ESP-E	44°34'4.800"N	67°17' 420"W
	ESP-F	44°42'11.736"N	66°29'8.244"W

Table S3. Maine Department of Marine Resources (DMR) blue mussel (*Mytilus edulis*) sampling stations used in this study.

Year	Toxin station	Latitude (dms)	Longitude (dms)
2014	York River	43°7'38.172"N	70°38'39.336"W
	Basin Point	43°44'18.528"N	70°2'36.348"W
	Potts Point	43°43'50.304"N	70°1'32.0160"W
	Head Beach	43°43'01.6"N	69°50'60.0"W
	Christmas Cove	43°50'51.216"N	69°33'25.272"W
	Port Clyde	43°55'30.936"N	69°15'32.400"W
2017	Gove Point	44°54'19.620"N	67°3'22.356"W
	Gleason Cove	44°58'15.0240"N	67°3'9.144"W
	Lubec Channel	44°51'32.0760"N	66°58'49.00800"W
2019	Gove Point	44°54'19.620"N	67°3'22.356"W
	Gleason Cove	44°58'15.0240"N	67°3'9.144"W
	Lubec Channel	44°51'32.0760"N	66°58'49.00800"W

Table S4. Model R^2 (Pearson's correlation coefficient squared) and free parameter values of the local fits of the 15 selected good fits (Figs. 3, 4, Supporting Information Tables S2-3). ESP: Environmental Sample Processor (locations, represented by letters; stations are indicated in Fig. 1); M0: model 0; M0L: model 0 lagged; M1: model 1; M2: model 2; M3: model 3; M4: model 4; SD: standard deviation; ns: non-significant difference; * $p \leq 0.05$; ** $0.001 < p \leq 0.01$; *** $p \leq 0.001$. β and γ : toxin uptake and depuration parameters; R^2 : Pearson's correlation coefficient squared.

Year	ESP	Mussel toxicity station	Distance between pairs (km)	M0 R^2	M0L R^2	M0L n° days	M1 β	M1 γ	M1 R^2	M2 R^2	M3 β	M3 R^2	M4 R^2
2014	A	York River	17	0.70*	0.89**	1	0.015	0.17	0.86**	0.83**	0.015	0.86**	0.86**
2014	B	Basin Point	7.8	0.35 ^{ns}	0.52 ^{ns}	2	0.054	0.33	0.67*	0.38 ^{ns}	0.024	0.51 ^{ns}	0.51 ^{ns}
2014	B	Potts Point	6.2	0.37 ^{ns}	0.57 ^{ns}	2	0.054	0.35	0.75*	0.37 ^{ns}	0.024	0.53 ^{ns}	0.53 ^{ns}
2014	C	Head Beach	33	0.96**	0.75 ^{ns}	1	0.066	0.13	0.96**	0.95**	0.072	0.96**	0.96**
2014	C	Christmas Cove	9.4	0.60 ^{ns}	0.65 ^{ns}	6	0.033	0.19	0.82*	0.68 ^{ns}	0.027	0.79*	0.78*

2014	C	Port Clyde	21	0.59 ^{ns}	0.82 [*]	7	0.012	0.03	0.90 [*]	0.83 [*]	0.024	0.74 ^{ns}	0.74 ^{ns}
2017	D	Gove Point	42	0.04 ^{ns}	0.72 ^{**}	7	0.009	0.09	0.91 ^{***}	0.90 ^{***}	0.012	0.90 ^{***}	0.90 ^{***}
2017	D	Gleason Cove	49	0.02 ^{ns}	0.81 ^{**}	7	0.03	0.11	0.99 ^{***}	0.93 ^{***}	0.039	0.98 ^{***}	0.93 ^{***}
2017	D	Lubec Channel	41	0.06 ^{ns}	0.87 ^{***}	2	0.021	0.15	0.94 ^{***}	0.88 ^{***}	0.021	0.94 ^{***}	0.94 ^{***}
2019	E	Gove Point	46	0.01 ^{ns}	0.69 [*]	7	0.021	0.09	0.77 ^{**}	0.78 ^{**}	0.03	0.58 [*]	0.57 [*]
2019	E	Gleason Cove	53	0.28 ^{ns}	0.42 ^{ns}	3	0.012	0.05	0.41 ^{ns}	0.42 ^{ns}	0.048	0.33 ^{ns}	0.16 ^{ns}
2019	E	Lubec Channel	46	0.97 ^{**}	0.98 ^{**}	2	0.027	0.15	0.98 ^{**}	0.96 ^{**}	0.027	0.98 ^{**}	0.98 ^{**}
2019	F	Gove Point	50	0.01 ^{ns}	0.78 ^{**}	1	0.021	0.13	0.97 ^{***}	0.97 ^{***}	0.021	0.95 ^{***}	0.95 ^{***}
2019	F	Gleason Cove	54	0.92 [*]	0.94 [*]	1	0.066	0.29	0.98 [*]	0.85 ^{ns}	0.027	0.91 [*]	0.89 ^{ns}
2019	F	Lubec Channel	43	0.93 ^{**}	0.91 [*]	4	0.045	0.17	0.76 ^{ns}	0.82 [*]	0.039	0.78 [*]	0.75 ^{ns}
		Overall/mean ($\pm$ SD)	35 ($\pm$ 17)	0.12 ^{***}	0.45 ^{***}	2	0.032 ($\pm$ 0.020)	0.162 ($\pm$ 0.095)	0.90 ^{***}	0.71 ^{***}	0.030 ($\pm$ 0.015)	0.85 ^{***}	0.69 ^{***}

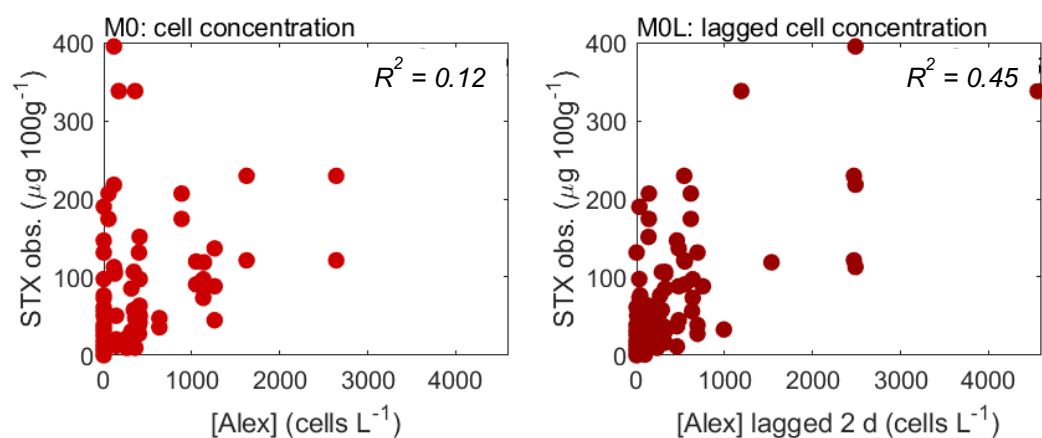


Fig. S1. Correlation plots of the observed mussel toxicity [μg saxitoxin (STX) equivalent 100g^{-1} soft tissue] and observed *Alexandrium catenella* (Alex) concentrations (cells L^{-1}) for the models M0 and M0L (Table 1 and Supporting Information Table S4). R^2 : Person's correlation coefficient squared.

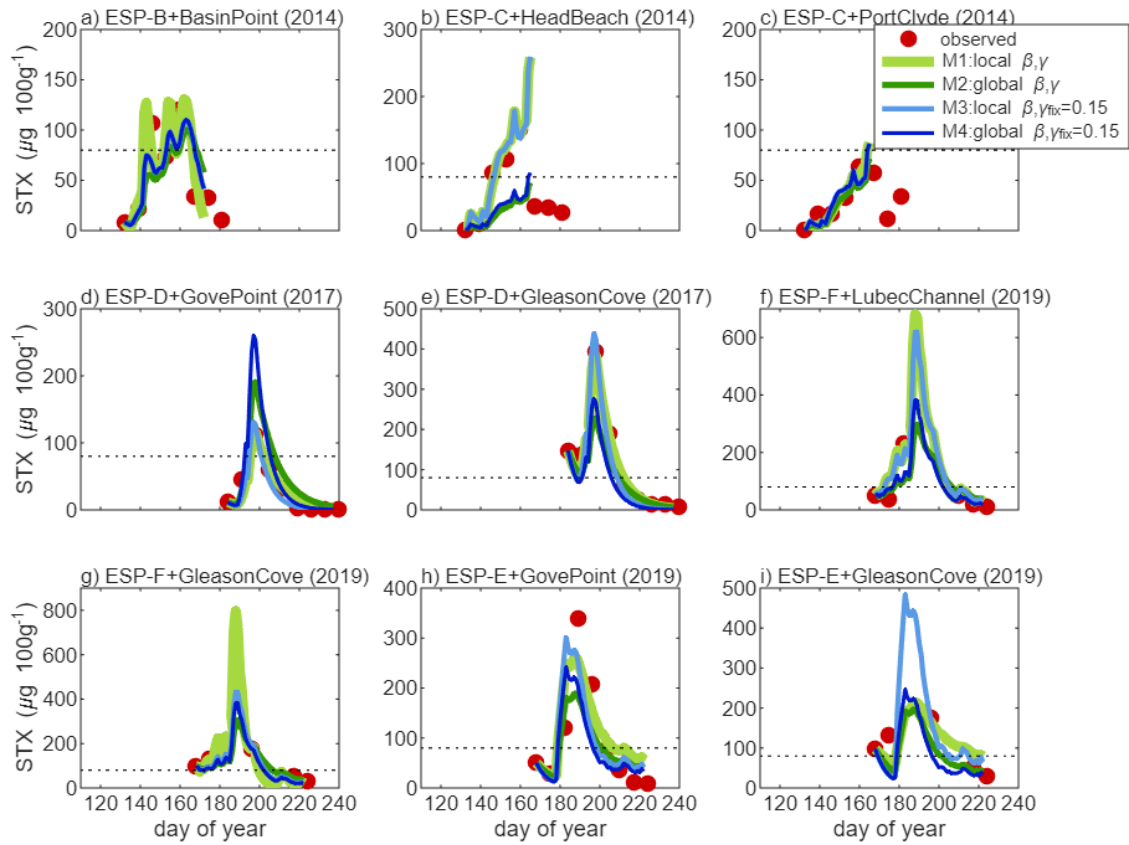


Fig. S2. Time series of observed paralytic shellfish toxin in blue mussel (*Mytilus edulis*) tissues [μg saxitoxin (STX) equivalents (eq.) 100g^{-1} soft tissue] sampled at coastal Maine Department of Marine Resources (DMR) stations (Fig. 1), and modeled toxicity (μg STX eq. 100g^{-1} soft tissue) (Table 1 and Supporting Information Table S4). Dashed line represents the regulatory limit (i.e., $80 \mu\text{g}$ STX eq. 100g^{-1} soft tissue). β : toxin uptake parameter; γ : toxin depuration parameters; ESP: Environmental Sample Processor; M1, M2, M3, and M4: models 1 through 4.